

Toxic proteins in neurodegenerative disease.

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A broad range of neurodegenerative disorders is characterized by neuronal damage that may be caused by toxic, aggregation-prone proteins. As genes are identified for these disorders and cell culture and animal models are developed, it has become clear that a major effect of mutations in these genes is the abnormal processing and accumulation of misfolded protein in neuronal inclusions and plaques. Increased understanding of the cellular mechanisms for disposal of abnormal proteins and of the effects of toxic protein accumulation on neuronal survival may allow the development of rational, effective treatment for these disorders.

Protein S-nitrosylation and oxidation contribute to protein misfolding in neurodegeneration.

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Neurodegenerative disorders like Alzheimer's disease and Parkinson's disease are characterized by progressive degeneration of synapses and neurons. Accumulation of misfolded/aggregated proteins represents a pathological hallmark of most neurodegenerative diseases, potentially contributing to synapse loss and neuronal damage. Emerging evidence suggests that misfolded proteins accumulate in the diseased brain at least in part as a consequence of excessively generated reactive oxygen species (ROS) and reactive nitrogen species (RNS). Mechanistically, not only disease-linked genetic mutations but also known risk factors for neurodegenerative diseases, such as aging and exposure to environmental toxins, can accelerate production of ROS/RNS, which contribute to protein misfolding - in many cases mimicking the effect of rare genetic mutations known to be linked to the disease. This review will focus on the role of RNS-dependent post-translational modifications, such as S-nitrosylation and tyrosine nitration, in protein misfolding and aggregation. Specifically, we will discuss molecular mechanisms whereby RNS disrupt the activity of the cellular protein quality control machinery, including molecular chaperones, autophagy/lysosomal pathways, and the ubiquitin-proteasome system (UPS). Because chronic accumulation of misfolded proteins can trigger mitochondrial dysfunction, synaptic damage, and neuronal demise, further characterization of RNS-mediated protein misfolding may establish these molecular events as therapeutic targets for intervention in neurodegenerative diseases.

Cell-to-cell transmission of pathogenic proteins in neurodegenerative diseases.

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A common feature of many neurodegenerative diseases is the deposition of β -sheet-rich amyloid aggregates formed by proteins specific to these diseases. These protein aggregates are thought to cause neuronal dysfunction, directly or indirectly. Recent studies have strongly implicated cell-to-cell transmission of misfolded proteins as a common mechanism for the onset and progression of various neurodegenerative disorders. Emerging evidence also suggests the presence of conformationally diverse 'strains' of each type of disease protein, which may be another shared feature of amyloid aggregates, accounting for the tremendous heterogeneity within each type of neurodegenerative disease. Although there are many more questions to be answered, these studies have opened up new avenues for

therapeutic interventions in neurodegenerative disorders.

Proteinopathies: Deciphering Physiology and Mechanisms to Develop Effective Therapies for Neurodegenerative Diseases.

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Neurodegenerative diseases (NDs) are a cluster of diseases marked by progressive neuronal loss, axonal transport blockage, mitochondrial dysfunction, oxidative stress, neuroinflammation, and aggregation of misfolded proteins. NDs are more prevalent beyond the age of 50, and their symptoms often include motor and cognitive impairment. Even though various proteins are involved in different NDs, the mechanisms of protein misfolding and aggregation are very similar. Recently, several studies have discovered that, like prions, these misfolded proteins have the inherent capability of translocation from one neuron to another, thus having far-reaching implications for understanding the processes involved in the onset and progression of NDs, as well as the development of innovative therapy and diagnostic options. These misfolded proteins can also influence the transcription of other proteins and form aggregates, tangles, plaques, and inclusion bodies, which then accumulate in the CNS, leading to neuronal dysfunction and neurodegeneration. This review demonstrates protein misfolding and aggregation in NDs, and similarities and differences between different protein aggregates have been discussed. Furthermore, we have also reviewed the disposal of protein aggregates, the various molecular machinery involved in the process, their regulation, and how these molecular mechanisms are targeted to build innovative therapeutic and diagnostic procedures. In addition, the landscape of various therapeutic interventions for targeting protein aggregation for the effective prevention or treatment of NDs has also been discussed.

A central role for calcineurin in protein misfolding neurodegenerative diseases.

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Accumulation of misfolded/unfolded aggregated proteins in the brain is a hallmark of many neurodegenerative diseases affecting humans and animals. Dysregulation of calcium (Ca^{2+}) and disruption of fast axonal transport (FAT) are early pathological events that lead to loss of synaptic integrity and axonal degeneration in early stages of neurodegenerative diseases. Dysregulated Ca^{2+} in the brain is triggered by accumulation of misfolded/unfolded aggregated proteins in the endoplasmic reticulum (ER), a major Ca^{2+} storing organelle, ultimately leading to neuronal dysfunction and apoptosis. Calcineurin (CaN), a Ca^{2+} /calmodulin-dependent serine/threonine phosphatase, has been implicated in T cells activation through the induction of nuclear factor of activated T cells (NFAT). In addition to the involvement of several other signaling cascades, CaN has been shown to play a role in early synaptic dysfunction and neuronal death. Therefore, inhibiting hyperactivated CaN in early stages of disease might be a promising therapeutic strategy for treating patients with protein misfolding diseases. In this review, we briefly summarize the structure of CaN, inhibition mechanisms by which immunosuppressants inhibit CaN, role of CaN in maintaining neuronal and synaptic integrity and homeostasis and the role played by CaN in protein unfolding/misfolding neurodegenerative diseases.

Autophagy Dysfunction in Alzheimer's Disease: Mechanistic Insights and New Therapeutic Opportunities.

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Alzheimer's disease (AD) is a neurodegenerative disorder characterized by progressive memory loss due to aberrant accumulation of misfolded proteins inside and outside neurons and glial cells, leading to a loss of cellular protein homeostasis. Today, no therapy is available to block or slow down AD progression, and the mechanisms of the disease are not fully understood. Autophagy is an intracellular degradation pathway crucial to maintaining cellular homeostasis by clearing damaged organelles, pathogens, and unwanted protein aggregates. In recent years, autophagy dysfunction has gained considerable attention in AD and other neurodegenerative diseases because it has been linked to the accumulation of misfolded proteins that ultimately causes neuronal death in many of these disorders. Interestingly, autophagy-activating compounds have also shown some promising results in both clinical trials and preclinical studies. This review aims at summarizing the current knowledge on autophagy dysfunction in the context of AD pathophysiology, providing recent mechanistic insights on AD-mediated autophagic flux disruption and highlighting potential and novel therapeutic opportunities that target this system for AD therapy.

A Specific Mini-Intrabody Mediates Lysosome Degradation of Mutant Huntingtin.

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Accumulation of misfolded proteins leads to many neurodegenerative diseases that can be treated by lowering or removing mutant proteins. Huntington's disease (HD) is characterized by the intracellular accumulation of mutant huntingtin (mHTT) that can be soluble and aggregated in the central nervous system and causes neuronal damage and death. Here, an intracellular antibody (intrabody) fragment is generated that can specifically bind mHTT and link to the lysosome for degradation. It is found that delivery of this peptide by either brain injection or intravenous administration can efficiently clear the soluble and aggregated mHTT by activating the lysosomal degradation pathway, resulting in amelioration of gliosis and dyskinesia in HD knock-in (KI-140Q) mice. These findings suggest that the small intrabody peptide linked to lysosomes can effectively lower mutant proteins and provide a new approach for treating neurodegenerative diseases that are caused by the accumulation of mutant proteins.

Targeting Neuroinflammation by Pharmacologic Downregulation of Inflammatory Pathways Is Neuroprotective in Protein Misfolding Disorders.

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Neuroinflammation plays a crucial role in the development of neurodegenerative protein misfolding disorders. This category of progressive diseases includes, but is not limited to, Alzheimer's disease, Parkinson's disease, and prion diseases. Shared pathogenesis involves the accumulation of misfolded proteins, chronic neuroinflammation, and synaptic dysfunction, ultimately leading to irreversible neuronal loss, measurable cognitive deficits, and death. Presently, there are few to no effective treatments to halt the advancement of neurodegenerative diseases. We hypothesized that directly targeting neuroinflammation by downregulating the transcription factor, NF- κ B, and the inflammasome protein,

NLRP3, would be neuroprotective. To achieve this, we used a cocktail of RNA targeting therapeutics (SB_NI_112) shown to be brain-penetrant, nontoxic, and effective inhibitors of both NF- κ B and NLRP3. We utilized a mouse-adapted prion strain as a model for neurodegenerative diseases to assess the aggregation of misfolded proteins, glial inflammation, neuronal loss, cognitive deficits, and lifespan. Prion-diseased mice were treated either intraperitoneally or intranasally with SB_NI_112. Behavioral and cognitive deficits were significantly protected by this combination of NF- κ B and NLRP3 downregulators. Treatment reduced glial inflammation, protected against neuronal loss, prevented spongiotic change, rescued cognitive deficits, and significantly lengthened the lifespan of prion-diseased mice. We have identified a nontoxic, systemic pharmacologic that downregulates NF- κ B and NLRP3, prevents neuronal death, and slows the progression of neurodegenerative diseases. Though mouse models do not always predict human patient success and the study was limited due to sample size and number of dosing methods utilized, these findings serve as a proof of principle for continued translation of the therapeutic SB_NI_112 for prion disease and other neurodegenerative diseases. Based on the success in a murine prion model, we will continue testing SB_NI_112 in a variety of neurodegenerative disease models, including Alzheimer's disease and Parkinson's disease.

Proteostasis and Ribostasis Impairment as Common Cell Death Mechanisms in Neurodegenerative Diseases.

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The cellular homeostasis of proteins (proteostasis) and RNA metabolism (ribostasis) are essential for maintaining both the structure and function of the brain. However, aging, cellular stress conditions, and genetic contributions cause disturbances in proteostasis and ribostasis that lead to protein misfolding, insoluble aggregate deposition, and abnormal ribonucleoprotein granule dynamics. In addition to neurons being primarily postmitotic, nondividing cells, they are more susceptible to the persistent accumulation of abnormal aggregates. Indeed, defects associated with the failure to maintain proteostasis and ribostasis are common pathogenic components of age-related neurodegenerative diseases, including Alzheimer's disease, Parkinson's disease, and amyotrophic lateral sclerosis. Furthermore, the neuronal deposition of misfolded and aggregated proteins can cause both increased toxicity and impaired physiological function, which lead to neuronal dysfunction and cell death. There is recent evidence that irreversible liquid-liquid phase separation (LLPS) is responsible for the pathogenic aggregate formation of disease-related proteins, including tau, α -synuclein, and RNA-binding proteins, including transactive response DNA-binding protein 43, fused in sarcoma, and heterogeneous nuclear ribonucleoprotein A1. Investigations of LLPS and its control therefore suggest that chaperone/disaggregase, which reverse protein aggregation, are valuable therapeutic targets for effective treatments for neurological diseases. Here we review and discuss recent studies to highlight the importance of understanding the common cell death mechanisms of proteostasis and ribostasis in neurodegenerative diseases.

Role of molecular chaperones in neurodegenerative disorders.

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Many major neurodegenerative diseases, including Amyotrophic Lateral Sclerosis, Alzheimer's disease, Parkinson's disease, Huntington Disease and other polyglutamine expansion disorders, are associated

with degeneration and death of specific neuronal populations due to accumulation of certain abnormal polypeptides. These misfolded species aggregate and form inclusion bodies and their neurotoxicity is associated with the aggregation. To handle a build-up of abnormal proteins cells employ a complicated machinery of molecular chaperones and various proteolytic systems. Chaperones facilitate refolding or degradation of misfolded polypeptides, prevent protein aggregation and play a role in formation of aggresome, a centrosome-associated body to which small cytoplasmic aggregates are transported. The ubiquitin-proteasome proteolytic system is critical for reducing the levels of soluble abnormal proteins, while autophagy plays the major role in clearing of cells from protein aggregates. Accumulation of the aggregation prone proteins activates signal transduction pathways that control cell death, including JNK pathway that controls viability of a cell in various models of Parkinson's and Huntington's diseases. The major chaperone Hsp72 can interfere with this signalling pathway, thus promoting survival. A very important consequence of a build-up and aggregation of misfolded proteins is impairment of the ubiquitin-proteasome degradation system and suppression of the heat shock response. Such an inhibition of the major cell defense systems may play a critical role in neurodegeneration. Here, it is suggested that these changes may reflect a senescence-like programme initiated by the aggregated abnormal polypeptides. Pathways that control the fate of misfolded proteins, for example molecular chaperones or proteolytic systems, may become interesting novel targets for therapy of neurodegenerative disorders.
